



SNAPDRAGON® AI LAB BUILD & PRESENT CHALLENGE

Qualcomm Snapdragon Smart PC Guardian

AI-powered monitoring, prediction and guidance for next-generation smart PCs

MONITOR • PREDICT • EXPLAIN • RECOMMEND

Rahul Das | B.Tech Electronics & Communication Engineering
Lovely Professional University

SHORT PITCH PRESENTATION

Powerful PCs are still hard to understand

The issue is not a lack of sensors. It is the lack of a reasoning layer that connects them.

**“My laptop is slow, hot,
and the battery is
dropping.
What is actually
wrong?”**

Today, students often open separate tools for CPU, GPU, RAM, battery and temperature - then interpret the numbers themselves.

USER PAIN

Signals that matter

- 1** CPU compute pressure
- 2** GPU graphics / AI load
- 3** RAM / VRAM memory pressure
- 4** THERMALS heat / throttling
- 5** BATTERY / POWER endurance / energy

THE GAP

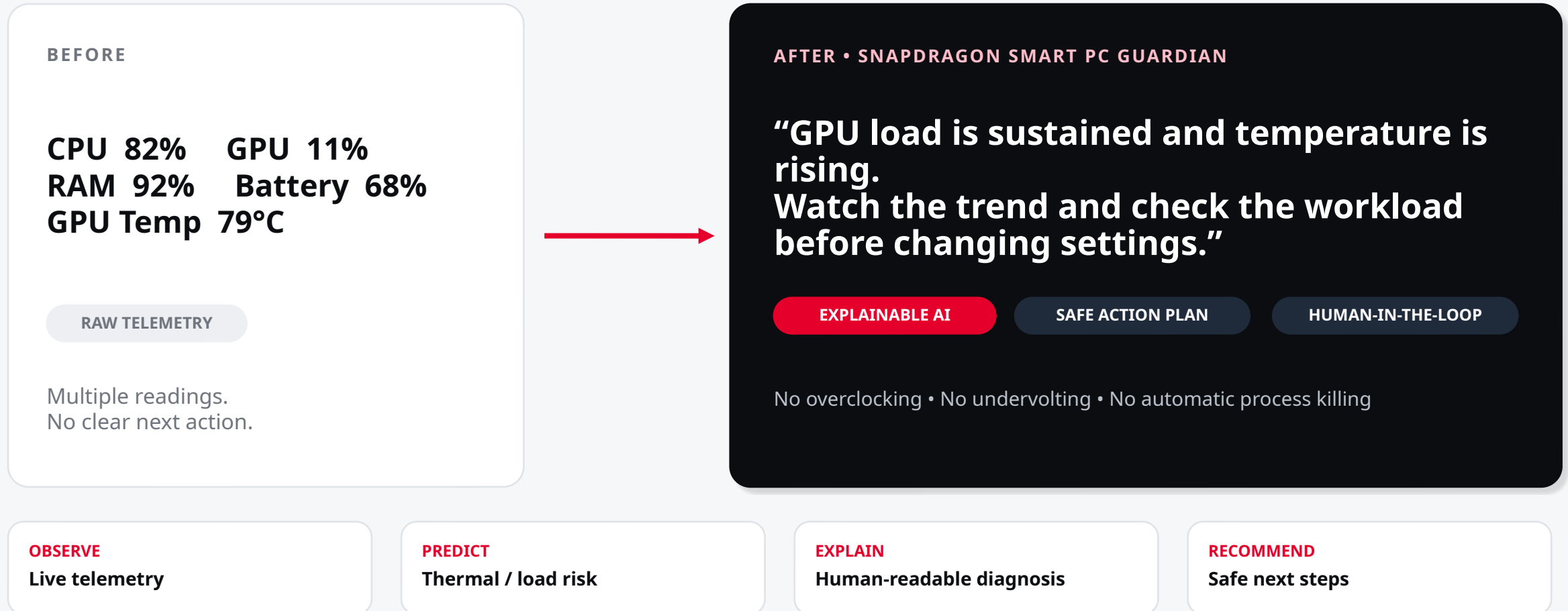
Numbers without context

Guardian turns fragmented telemetry into a single answer:

- 1.** What is happening?
- 2.** Why does it matter?
- 3.** What should I check next?

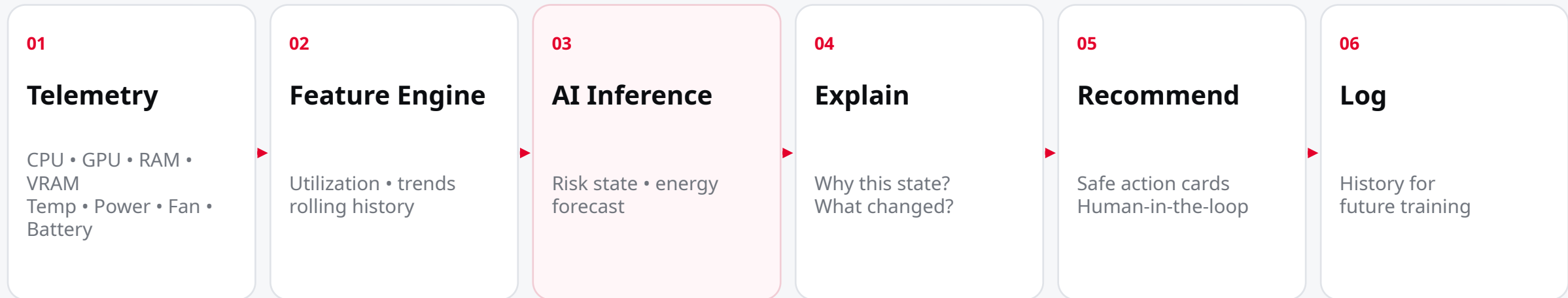
One intelligence layer for the whole PC

Guardian observes the machine like an engineer, interprets it like an AI analyst, and explains it in student-friendly language.



Observe → Contextualize → Predict → Explain → Recommend → Log

A modular pipeline keeps the experience simple while leaving room for Snapdragon CPU/GPU/NPU integration.



SNAPDRAGON ADAPTER

Platform-aware mapping

CPU / GPU / NPU telemetry can be surfaced through Windows and Qualcomm-specific interfaces when supported.

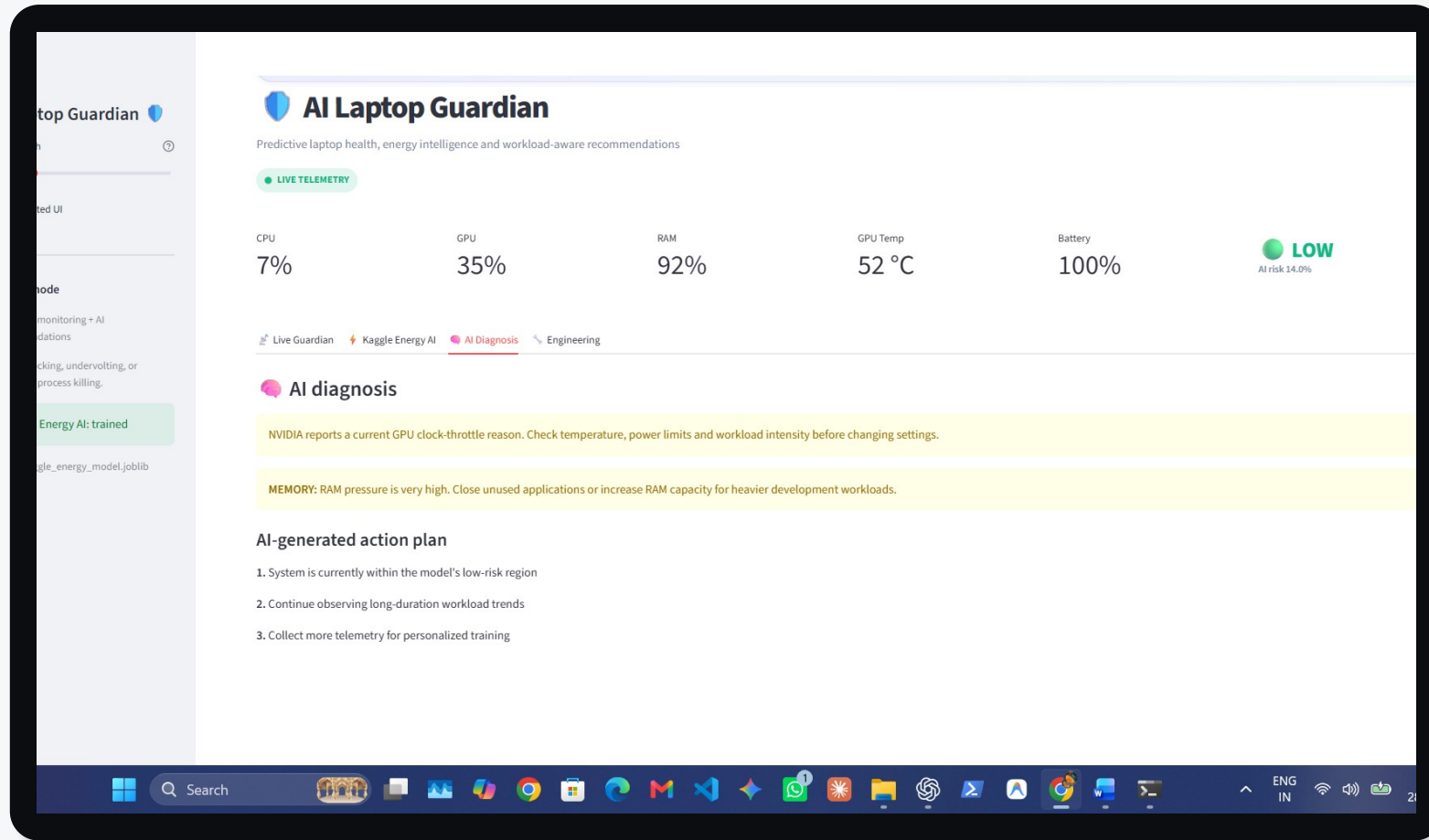
Windows on Arm: native vs emulated vs driver-dependent guidance

Design principle: recommend first, change second.

The user stays in control.

A control-room dashboard, not a spreadsheet

The working prototype combines live telemetry, a risk gauge, AI diagnosis, action planning and history.



01 LIVE KPIS

CPU • GPU • RAM • Temp • Battery

02 PREDICTIVE RISK

Low / Medium / High risk gauge

03 AI DIAGNOSIS

Plain-language explanation

04 ACTION PLAN

Safe next steps, no autonomous tuning

“The laptop does not just show data. It explains the data.”

Two AI tracks, kept technically honest

Laptop thermal intelligence and the Kaggle energy experiment are deliberately separated.

TRACK A

Laptop Risk Model

Prototype features

- 1 CPU / GPU utilization
- 2 RAM / VRAM pressure
- 3 Temperature / power
- 4 Battery state / trends

RESEARCH NEXT STEP

Train on real labeled laptop telemetry.

TRACK B • KAGGLE ENERGY EXPERIMENT

20-run model selection

20

CANDIDATE RUNS

0.6047

TEST R²

26.14 Wh

TEST MAE

57.20

RMSE

Best family: HistGradientBoosting • 19,735 rows • train/val/test split

This model predicts household/appliance energy demand - it is not a laptop wattmeter or thermal model.

Actual training run

```

val: testval: 1.0000000000000000
[19/20] Training split seed: 0
val: testval: 1.0000000000000000
[20/20] Training split seed: 0
val: testval: 1.0000000000000000
Refitting best configuration: hist_gbr_seed0

=== KAGGLE ENERGY AI: FINAL RESULT ===
Runs: 20
Best model: hist_gbr_seed0
Runs: 19,735
Train/Val/Test: 15,666 / 2,037 / 2,032
MAE: 26.14 Wh
RMSE: 57.20 Wh
R2: 0.6047

Save: C:\Users\rahul\Downloads\AI_Laptop_Guardian_FINAL_MORNING\val_laptop_guardian_model\kaggle_energy_model.joblib
NOTE: This model predicts household/appliance energy, not laptop power, temperature, or throttling.
All training is done on the Kaggle dataset.
Model: hist_gbr_seed0
Training completed successfully.
Press any key to continue.
PS C:\Users\rahul\Downloads\AI_Laptop_Guardian_FINAL_MORNING\val_laptop_guardian> .\UNW.bat
  
```

Built around real student workflows

The same intelligence layer can support daily laptop use now and evolve toward Snapdragon AI PCs.

AI / ML

Watch compute pressure during training or inference.

PROGRAMMING

Spot IDE, compiler and browser resource contention.

CLASSES

Understand sustained camera / browser / AI load.

ENGINEERING

Detect long simulation and graphics stress.

CONTENT + GAMING

Track GPU / VRAM / temperature together.

TRAVEL

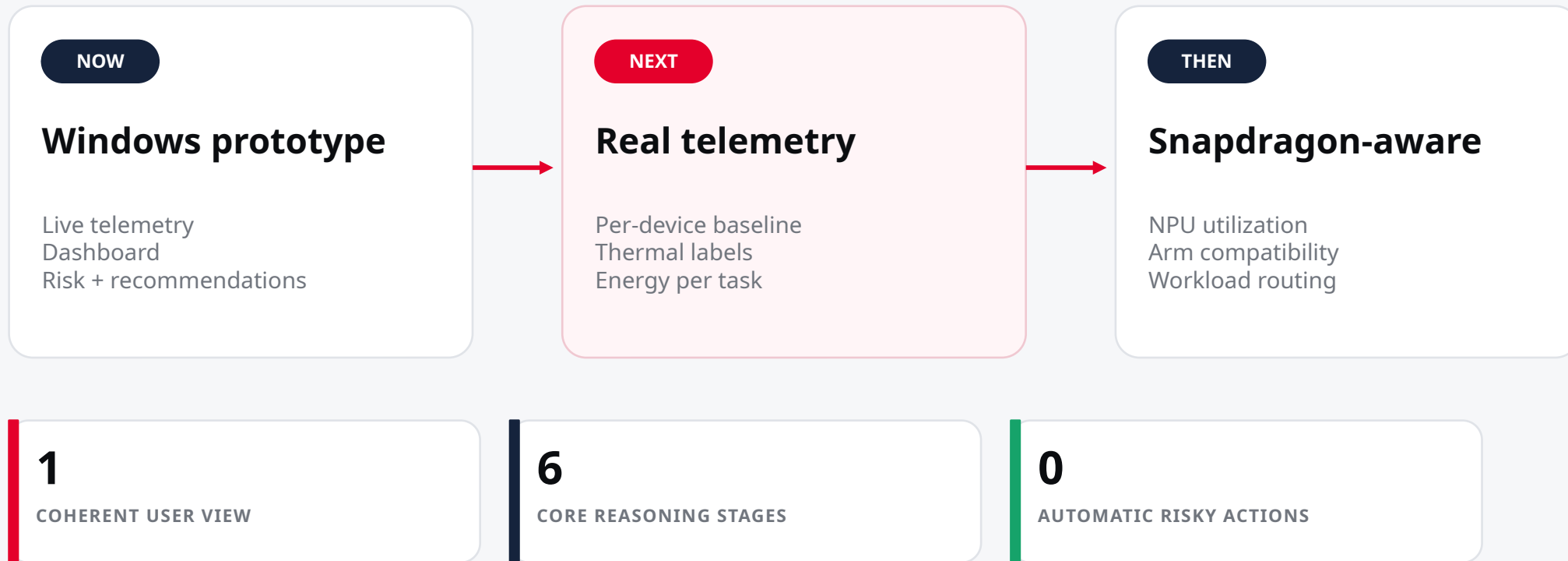
Balance performance against battery endurance.

WHY SNAPDRAGON AI PCs? Heterogeneous compute CPU + GPU + NPU roles **Windows on Arm** native / emulated / driver-aware context

Future: NPU-aware workload panel + Arm compatibility assistant

From prototype to a Snapdragon-first daily companion

The architecture is modular today and leaves a clear path toward real-device validation.



Value to students: less guesswork, clearer feedback, and a practical way to learn AI + systems

“Watch the laptop like an engineer. Explain it like a teacher. Recommend like an assistant.”